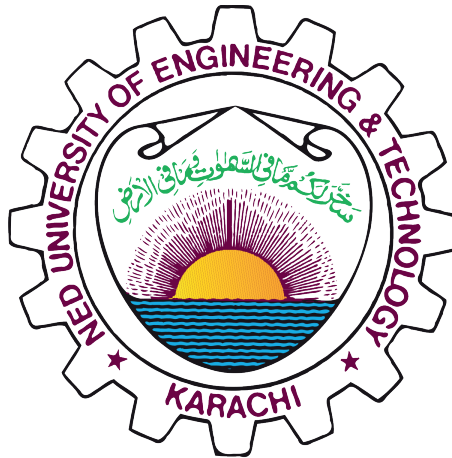




CS-327

**COMPUTER COMMUNICATION
NETWORKS**

ASSIGNMENT

SUBMITTED BY:

CS-23055

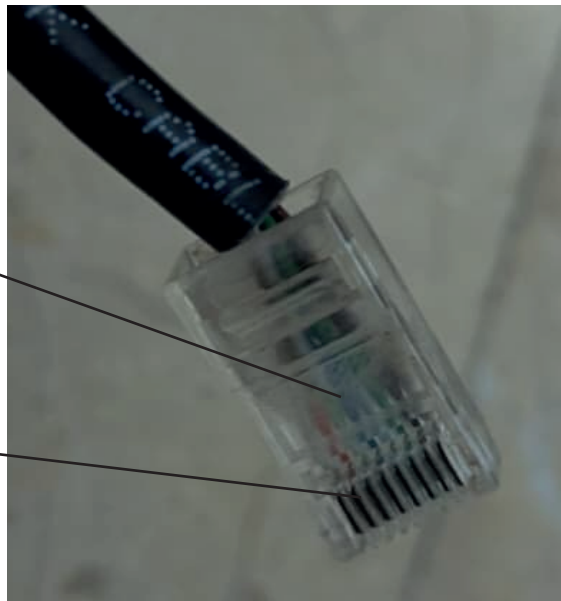
Q1: Identify and illustrate the type of physical media used for your home internet connection (attach snapshot of the said connection as well). List all the communication links available within your home network setup.

1. The internet connection in my home coming from the ISP uses a Cat5e Ethernet cable (RJ-45 connector) over twisted pair copper wiring which is plugged into the router.

Then my PC also connects through a Cat5e Ethernet cable (RJ-45, twisted pair copper) that runs from the router to the PC.

Twisted Pair Cable
(Inside)

RJ-45 connector



2. Television Cable: The TV Cable in my home uses a coaxial cable which connects the service provider directly to the TV set-top box, delivering digital TV channels.

Carries the signal

Outer shield for
interference prevention



Communication Links Available in My Home Network:

- Wi-Fi: Router connecting to mobile, laptop and TV
- Bluetooth: Mobile connecting to speaker, headphone
- Ethernet Cable(LAN): Router connecting to Desktop PC
- Coaxial Cable: Service Provider to TV set-top box.
- USB: PC connecting to Peripherals
- HDMI: CPU connecting to Monitor

Q2: Explain at least five commonly used network services (e.g., web, email), describe their purpose.

- **HTTP/HTTPS:**

Protocols used by web browsers and servers to communicate and exchange data.

Purpose: To deliver websites, web pages, and APIs securely to users.

- **Cloud Storage:**

Cloud storage allows users to save data on remote servers that are accessible via the internet, instead of local devices.

Purpose: To store, back up, and access files online from anywhere.

- **Security Services:**

These services include firewalls and antivirus software that protect systems from threats like viruses, malware, and unauthorized access.

Purpose: To ensure system security and protect data from cyber threats.

- **iPTV/Streaming Service:**

iPTV (Internet Protocol Television) uses the internet to stream live TV and on-demand content instead of using traditional broadcast methods.

Purpose: To provide digital television and video streaming over internet connections.

- **VPN Service:**

A VPN (Virtual Private Network) creates a secure tunnel between your device and the internet, encrypting your data and hiding your IP address.

Purpose: To ensure privacy, secure internet access, and allow remote connection to private networks.

Q3: Analyze and compare at least three server and three desktop machines from different vendors (e.g., Dell, HP, Lenovo) based on processor, RAM, storage, networking, pricing, and highlight any unique or sustainable features.

1) Features	Dell Slim Desktop (ECS1250) (Base Model)	Dell PowerEdge T160 Tower Server
Processor	Intel® Core™ i3 14100, 4 cores	One Intel® Xeon® 6300 or Xeon E-2400 series, or Pentium G7400
Memory (RAM)	8GB, 8GBx1, DDR5, 4800MT/s; 2 slots, supporting upto 64GB	4 × DDR5 ECC UDIMM slots; up to 128 GB ECC RAM
Storage	512GB M.2 PCIe NVMe Solid State Drive	Up to 3 × 3.5" + 2 × 2.5" drive bays for SAS/SATA, max ~63 TB total storage
Networking	1 RJ45 (1 Gbps) Ethernet port	2 x 1GbE LOM(LAN on Motherboard)
Price	Starting at \$399	Starting at \$1069

2) Features	HP OmniDesk Desktop M02-0055t PC	HPE ProLiant ML110 Gen11 (HP Enterprise)
Processor	Intel® Core™ i5-14400 (up to 4.7 GHz with Intel® Turbo Boost Technology, 20 MB L3 cache, 10C, 16T)	4th Gen and 5th Gen Intel® Xeon® Scalable Processors (8-32C), 3.2Ghz Max, 22.5 MB L3 – 60 MB L3
Memory (RAM)	16 GB DDR5-4800 MT/s (1 x 16 GB) 2 DIMM slots	DDR 5 1536 GB with 96 GB DDR5 across 16 DIMM slots
Storage	512 GB PCIe® Gen4 NVMe™ M.2 SSD	between 4 to 8 LFF (3.5-inch) SAS/SATA drives or 8 to 16 SFF (2.5-inch) SAS/SATA drives
Networking	Integrated 10/100/1000 GbE LAN	2× 1 GbE NICs built-in
Price	Starting at \$589.99	Starting at \$1,855.86

3) Features	ThinkCentre M70t Gen 6 (Intel) Tower	ThinkSystem ST650 V3 Tower Server
Processor	Intel® Core™ Ultra 7 265 vPro® Processor	Latest 5th Gen Intel® Xeon® Scalable processors
Memory (RAM)	4 x UDIMM Slots supporting upto 128 GB	Up to 4TB TruDDR5 memory in 5600MHz across 32 DIMM slots
Storage	Up to 2TB M.2 Gen4 SSD 2TB 7200rpm HDD 3.5" SATA	Up to 16x 3.5" HS SAS/SATA (up to 16x optional NVMe SSDs) or Up to 32x 2.5" HS SAS/SATA (up to 24x optional NVMe SSDs)
Networking	Ethernet (RJ45)	2x 10GbE ports; 1x 1GbE dedicated management port
Price	\$1,129.00	Starting at \$3,651.53

Unique or sustainable features:

- Servers come with way more upgradable options than Desktops.
- Desktops usually maxes out at 128GB of RAM while Servers supports multiple TBs of RAM.
- Servers are designed for energy-efficient 24/7 operation with features like redundant power supplies and virtualization support, while desktops focus on power efficiency for individual use, making servers more sustainable for handling multiple workloads and desktops more sustainable for single-user tasks.

Q4: List down host devices (10)

- Desktop
- Laptop
- Smartphone
- Smart TV
- Printer (Network Enabled)
- Smartwatch
- Wi-Fi Router
- Gaming Consoles
- CCTV (Wifi) Camera
- ATM Machine